

Regression Analysis

Bernd Skiera, Jochen Reiner, and Sönke Albers

Contents

Introduction	300
Statistical Explanation of the Method	300
Problem Statement	300
Objective Function and Estimation of Regression Coefficients	301
Goodness of Fit	304
Significance Testing	305
Standardization of Coefficients	306
Interpretation of Results	307
Results of Numerical Example	307
Assumptions	309
Procedure	310
Efficiency of Estimators	311
Test for Multicollinearity	311
Test for Autocorrelation	313
Test for Heteroscedasticity	315
Identification of Outliers	317
Transformation of Variables	318
Implications of the Analysis	320
Endogeneity	321
Further Topics	324
Software	325
Summary	326
References	326

B. Skiera (✉) · J. Reiner
Goethe University Frankfurt, Frankfurt, Germany
e-mail: skiera@skiera.de; jreiner@wiwi.uni-frankfurt.de

S. Albers
Kuehne Logistics University, Hamburg, Germany
e-mail: soenke.albers@the-klu.org

Abstract

Linear regression analysis is one of the most important statistical methods. It examines the linear relationship between a metric-scaled dependent variable (also called endogenous, explained, response, or predicted variable) and one or more metric-scaled independent variables (also called exogenous, explanatory, control, or predictor variable). We illustrate how regression analysis work and how it supports marketing decisions, e.g., the derivation of an optimal marketing mix. We also outline how to use linear regression analysis to estimate nonlinear functions such as a multiplicative sales response function. Furthermore, we show how to use the results of a regression to calculate elasticities and to identify outliers and discuss in details the problems that occur in case of autocorrelation, multicollinearity and heteroscedasticity. We use a numerical example to illustrate in detail all calculations and use this numerical example to outline the problems that occur in case of endogeneity.

Keywords

Regression analysis · Marketing mix modeling · Elasticities · Multicollinearity · Autocorrelation · Outlier detection · Endogeneity · Sales response function

Introduction

Linear regression analysis is one of the most important statistical methods. It examines the linear relationship between a metric-scaled dependent variable (also called endogenous, explained, response, or predicted variable) and one or more metric-scaled independent variables (also called exogenous, explanatory, control, or predictor variable). Often, the dependent variable describes the success of marketing and the independent variables the factors that explain this success. Variables that describe the success of marketing are either “hard” success factors such as profit, sales volume (here referred to as quantity), and market share or “soft” success factors such as customers’ attitudes, purchase intention, and satisfaction. Variables that influence this success are frequently marketing instruments such as price, product, distribution, and communication. To illustrate how regression analysis works, we focus here on a numerical example with a hard success factor, namely, the estimation of quantity (i.e., sales volume), which is explained by variables such as advertising budget, price, and number of salespersons.

Statistical Explanation of the Method**Problem Statement**

We illustrate the basic idea of linear regression analysis by applying it to data from a (fictitious) company (displayed in Table 1). Their analysis should address the following business problems:

Table 1 Distribution of quantity and marketing instruments across districts

District	Quantity	Salespersons	Price	Advertising	Number of mailings
1	81,996	7	49	228,753	7,106
2	91,735	5	46	370,062	4,733
3	70,830	4	50	297,909	3,734
4	101,192	6	45	271,884	6,152
5	78,319	6	51	299,919	5,734
6	105,369	7	47	367,644	6,640
7	68,564	3	47	241,362	3,115
8	95,523	7	46	244,575	6,859
9	88,834	7	49	296,100	6,905
10	89,511	5	46	372,498	5,142
11	107,836	6	45	359,511	6,196
12	83,310	7	50	324,837	6,801
13	67,817	4	50	288,303	3,965
14	59,207	6	54	289,470	5,830
15	81,410	6	52	363,501	6,124
16	71,431	3	46	361,974	2,509
17	119,000	3	45	250,000	—

- What is the optimal price?
- What is the optimal budget for advertising and the sales force?

The data in Table 1 shows quantity at one point in time for 16 districts that were randomly selected from a much larger number of districts in which the company operates. The districts differ only in terms of price, advertising budget, and salespersons. For the moment, we ignore district 17 and the marketing instrument “number of mailings.”

The company has variable costs per unit sold of \$30 and costs for each salesperson of \$120,000 per year. Moreover, we assume that thus far the company has managed its marketing activities naively and has not systematically selected the value for each of its marketing instruments in each district. With this assumption, we exclude the problem of endogeneity (we do address this topic subsequently).

Objective Function and Estimation of Regression Coefficients

Linear regression analysis investigates the effect of metric-scaled independent variables (here, person, price, and advertising budget) on a metric-scaled dependent variable (here, quantity). We consider each district as one observation. The corresponding regression equation for the linear regression analysis is as follows:

$$y_i = b_0 + \sum_{k \in K} b_k \cdot x_{i,k} + e_i \quad (i \in I), \quad (1)$$

where

y_i = value of the i^{th} observation for the dependent variable,
 b_0 = intercept of the regression,
 b_k = regression coefficient capturing the influence of the k^{th} independent variable,
 $x_{i,k}$ = value of the i^{th} observation for the k^{th} independent variable,
 e_i = residual of the i^{th} observation,
 K = index set of the independent variables, and
 I = index set of the observations.

We can observe values of the dependent variable y_i and independent variables $x_{i,k}$ (see the respective values in Table 1), but we need to estimate the coefficients of the regression equation (Eq. 1), b_0 and b_k ($k \in K$), and the resulting residuals e_i ($i \in I$) (sometimes also referred to as “disturbance” or “error term”). The residual e_i describes the deviation between the observed value of the dependent variable y_i for the i^{th} observation and the predicted value for the dependent variable $\hat{y}_i$. The predicted value of the dependent variable is based on the estimated regression coefficients b_k and the respective observed independent variables $x_{i,k}$, as illustrated in Eq. 2:

$$\hat{y}_i = b_0 + \sum_{k \in K} b_k \cdot x_{i,k} \quad (i \in I). \quad (2)$$

The upper part of Fig. 1 illustrates the deviations of the observed values from the predicted values (for quantity, depending on advertising budget), with the predicted values lying on the regression line.

The aim of linear regression analysis is to estimate the coefficients of the regression equation b_0 and b_k ($k \in K$) so that the sum of the squared residuals (i.e., the sum over all squared differences between the observed values of the i^{th} observation of y_i and the corresponding predicted values $\hat{y}_i$) is minimized. The lower part of Fig. 1 illustrates this approach, which is called the “least squares method” (Stock and Watson 2015, p. 162). Simply speaking, the least squares method aims to minimize the sum of all rectangular areas displayed in the lower part of Fig. 1. The approach results in the following objective function:

$$\sum_{i \in I} e_i^2 = \sum_{i \in I} (y_i - \hat{y}_i)^2 = \sum_{i \in I} \left(y_i - b_0 - \sum_{k \in K} b_k \cdot x_{i,k} \right)^2 \rightarrow \min! \quad (3)$$

Using squared residuals is advantageous in that larger residuals receive a greater weight than smaller residuals and the solution for the objective function (Eq. 3) is algorithmically easy to determine (Pindyck and Rubinfeld 1998, p. 5). We do not derive the solution for the objective function (Eq. 3) here because a wide range of software (such as those presented in section “Software”) can perform this operation.

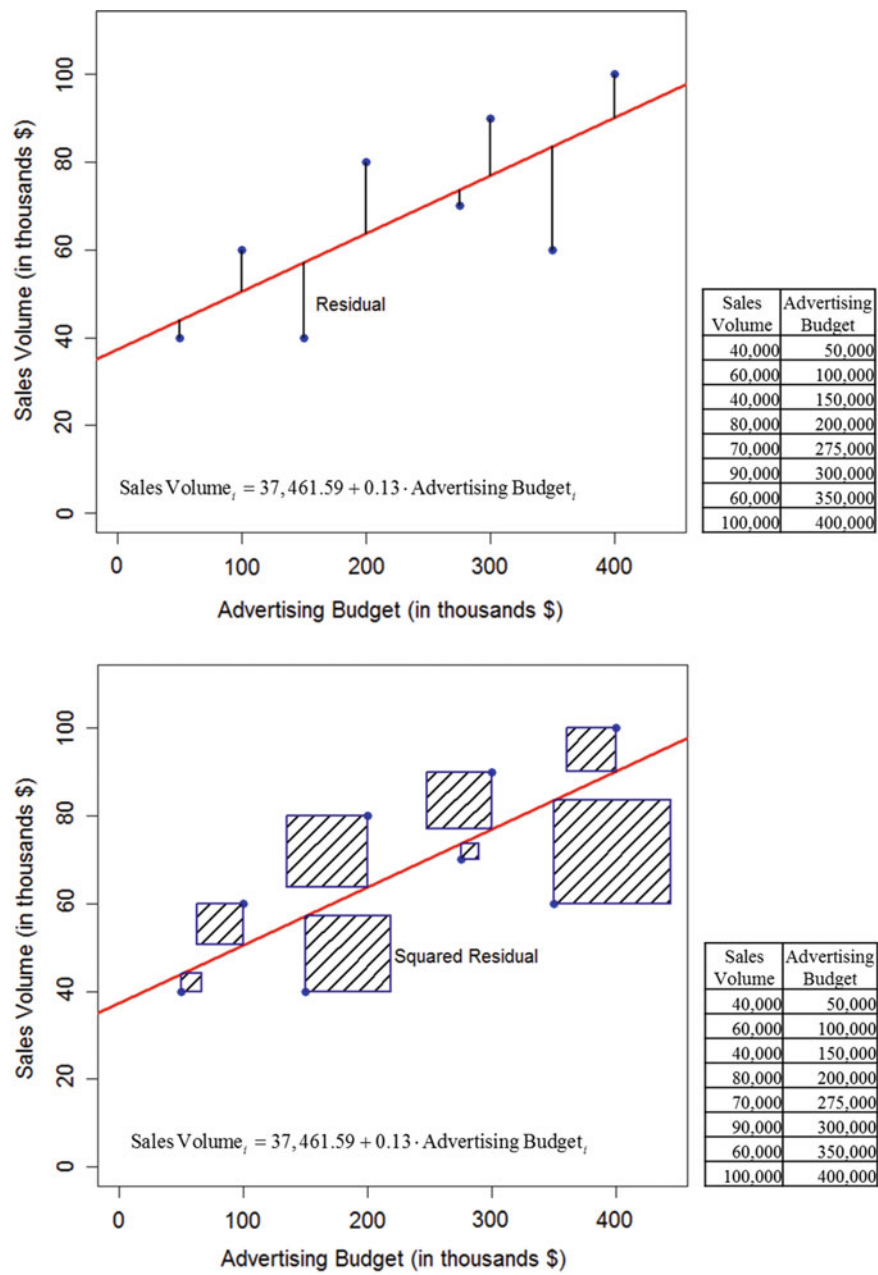


Fig. 1 Regression line with residuals and squared residuals

For more details on deriving the solution, see Wooldridge (2009, p. 27) and Gujarati (2003, p. 58), among many others.

Goodness of Fit

The assessment of goodness of fit for a linear regression analysis builds on the idea that the best estimate of the predicted value of a dependent variable is its mean value if nothing is known about the independent variables. Thus, a linear regression analysis is only meaningful if it can explain deviations from this mean value. More precisely, we determine goodness of fit by evaluating the extent to which considering the independent variables improves the “simple” prediction of just taking the mean value of the dependent variable. The coefficient of determination, called R^2 (see Eq. 4), measures goodness of fit by calculating the proportion of explained variance from the regression equation compared with the variance indicated by simply taking the mean value $\bar{y}$:

$$R^2 = \frac{\sum_{i \in I} (\hat{y}_i - \bar{y})^2}{\sum_{i \in I} (y_i - \bar{y})^2} \quad (4)$$

Thus, the linear regression analysis always explains at least the variance of the simple approach in that even if all independent variables have no explanatory power (the extreme worst case), the intercept captures the mean value of the dependent variable; in other words, the linear regression analysis reduces to simply predicting the mean value. Fortunately, adding one or more independent variables frequently improves this prediction. At the other extreme, the improvement in prediction might be so large that the predicted values of the dependent variable exactly match their observed values. In this case, the regression equation explains the total variance of the simple approach. R^2 can therefore take values between 0% and 100%. Negative values for R^2 can only arise when the researcher estimates the regression equation without an intercept, which is possible with most statistical programs; however, without an intercept, the linear regression analysis does not necessarily explain the variance of the simple approach.

These considerations should also make clear that including an additional independent variable never leads to a reduction of R^2 , because the explanatory power of the additional variable is at least zero, even if that variable explains nothing. In the extreme case that the number of observations corresponds to the number of estimated coefficients (more precisely, the intercept b_0 plus the number of regression coefficients b_k), we would essentially estimate a linear system of equations. The estimation of such a linear system always results in $R^2 = 100\%$, its largest possible value. To account for the fact that R^2 can only increase and does not penalize the use of variables that have no explanatory power, researchers frequently use the adjusted R^2 (R^2_{adj}), defined as follows (Wooldridge 2009, p. 200):

$$R^2_{adj} = R^2 - \frac{|K| \cdot (1 - R^2)}{|I| - |K| - 1}, \quad (5)$$

where

$|I|$ = number of elements in the index set of observations (equivalent to the number of observations) and

$|K|$ = number of elements in the index set of independent variables (equivalent to the number of regression coefficients).

Note that both numerator and denominator of the fraction are positive; therefore, the adjusted R^2 has at best the same value as the (unadjusted) R^2 . Large differences indicate that the regression equation contains many independent variables that do not explain the dependent variable (i.e., they have no explanatory power).

Significance Testing

Data are usually available only for a sample and not for an entire population, whether in the context of a large amount of surveys carried out by market research companies or data collected via panels and laboratory or field experiments. In the same vein, the presented 16 districts represent a (randomly selected) sample. In such a situation, it is necessary to determine the extent to which the results drawn from the sample also apply for the entire population. To do so, we must make an assumption about the distribution of the residuals.

For regression analysis (and many other methods), it is common to assume a normal distribution for the residuals. This assumption is based on the central limit theorem, which states that the mean from a random sample for any population (with finite variance), when standardized, has an asymptotic standard normal distribution (Wooldridge 2009, p. 758).

Using this assumption, we can conduct significance tests for the regression coefficients by calculating the error probability (often abbreviated as p-value) that the regression coefficients are nonzero. The significance test compares the determined error probability with a predetermined level of significance (often 5%). If the error probability is less than this predetermined level of significance, then the effect is considered significant.

To test the overall significance of a regression – that is, that all regression coefficients of the independent variables are 0 ($b_1 = b_2 = \dots = b_k = 0$) – we conduct the following F-Test with $|K|$ and $(|I| - |K| - 1)$ degrees of freedom:

$$F_{emp} = \frac{\frac{R^2}{|K|}}{\frac{1 - R^2}{|I| - |K| - 1}}. \quad (6)$$

If the F-value (F_{emp}) computed from Eq. 6 exceeds the critical F-value from the F table at a predetermined level of significance (often, as mentioned above, 5%),

the researcher can reject the null hypothesis that all independent variables have no effect on the dependent variable. Alternatively, many statistical software programs immediately report the error probability (p-value) for the F-value. If the error probability is less than the predetermined level of significance, the researcher can also reject the null hypothesis.

Whereas the F-test verifies the existence of a significant relationship between all independent variables and the dependent variable, the t-test determines the significance of each individual regression coefficient separately. Thus, it tests whether the null hypothesis of the coefficient being equal to 0 cannot be rejected with a certain error probability, frequently 5%. The t-test has $(|I| - |K| - 1)$ degrees of freedom and is calculated as follows:

$$t_{k, emp} = \frac{b_k}{s_k} \quad (k \in K), \quad (7)$$

where

$t_{k, emp}$ = empirical t-value for the k^{th} regression coefficient and
 s_k = estimated standard error of the k^{th} regression coefficient.

If the empirical t-value exceeds the critical t-value (from the t-distribution) at the chosen level of significance or, alternatively, the error probability (p-value) is less than the predetermined level of significance, then the coefficient is not equal to 0, and the impact of the k^{th} independent variable is significant.

Standardization of Coefficients

Often, it is necessary to compare the importance of the independent variables against one another. However, directly comparing the regression coefficients is hardly meaningful because the independent variables usually have different orders of magnitude (e.g., in Table 1, the advertising budget is measured in dollar values and person is measured in number of salespersons). The standardized regression coefficients beta_k allow for a meaningful comparison. They are calculated by multiplying the (unstandardized) regression coefficients b_k with the standard deviation σ_{xk} of the associated independent variable and dividing the result by the standard deviation of the dependent variable σ_y :

$$\text{beta}_k = b_k \cdot \frac{\sigma_{xk}}{\sigma_y} \quad (k \in K). \quad (8)$$

If the dependent and independent variables are standardized before running the regression analysis, then the regression coefficients beta_k and b_k will be the same. Thus, comparing the absolute values of all standardized regression coefficient beta_k shows the importance of the influence of the individual independent variables: Higher absolute values indicate a stronger influence.

Note, however, that using the standardized regression coefficient is discouraged when the standard deviations of the independent variables can be influenced. For example, if a company varied its product price to a greater extent than its advertising, then the standardized regression coefficients will indicate a stronger influence of the price due to the high standard deviation of the price (see Eq. 8). Therefore, it is preferable to calculate the elasticities of the independent variables, which are dimensionless and measure the responsiveness of the dependent variable to a change in one of the independent variables. Stated differently, they measure the percentage change of the dependent variable that corresponds to a 1% change of the independent variable. In the case of a linear regression, the elasticity is defined as follows:

$$\varepsilon_{y, x_k} = \frac{\frac{\partial y}{\partial x_k}}{\frac{y}{x_k}} = \frac{\partial y}{\partial x_k} \cdot \frac{x_k}{y} = b_k \cdot \frac{x_k}{y} \quad (k \in K). \quad (9)$$

Although the elasticity derived from Eq. 9 varies with the value of the independent variables, the common approach to calculate the elasticity for the independent variable x_k is to use the mean values of the independent variable x_k and the dependent variable y .

Interpretation of Results

Substantive insights should guide the interpretation of the results, rather than simply statistical criteria. In other words, researchers should first consider whether the regression equation captures all relevant variables via a meaningful functional relationship and then examine whether the signs of the regression coefficients are plausible. For the company in our example, the regression coefficients for advertising and person should be positive and negative, respectively, with regard to price. Then, to assess the size of the regression coefficients, calculating elasticities is often helpful. For example, advertising elasticities with values greater than 1 usually make little sense. The same holds for price elasticities with absolute values smaller than 1. Next to evaluating the substantive criteria, researcher should inspect statistical criteria such as the R^2 , the overall significance of the regression equation (F-test), the significance of the regression coefficients (t-test), and the subsequent assumptions for the least squares method.

Results of Numerical Example

In this section, we use the software program *R* to estimate the regression equation (codes for *R* but also other software programs such as STATA and SPSS are available on the authors' website). Figure 2 displays the result for this linear regression analysis with three marketing instruments (person, price, and advertising budget as independent variables and quantity as the dependent variable).

```

Call:
lm(formula = Quantity ~ Person + Price + Advertising, data = regdata)

Residuals:
    Min       1Q   Median       3Q      Max
-7650.1 -3383.9   287.8  3211.0  5047.8

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 210159.444  23729.909   8.856  0.000 ***
Person       6723.478   840.997   7.995  0.000 ***
Price      -3832.503   444.013  -8.632  0.000 ***
Advertising    0.069    0.024   2.903  0.013 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4546 on 12 degrees of freedom
Multiple R-squared:  0.9189,    Adjusted R-squared:  0.8986
F-statistic: 45.29 on 3 and 12 DF,  p-value: 8.077e-07

-----
"standardized coefficients - beta"
      Person      Price      Advertising
      0.665      -0.725       0.242
-----

Elasticities, at average values (not provided by R directly)
      Person      Price      Advertising
      0.446      -2.206       0.257

```

Fig. 2 Results of the linear regression analysis

The number of observations (here 16) is equal to the sum of the degrees of freedom (here 12) and the number of estimated coefficients (here 4). The R^2 ("R-squared") and the adjusted R^2 ("Adjusted R-squared") have values of 91.9% and 89.9%, respectively. The error probabilities determined for the F-test ("p-value") and the t-tests (column "Pr(>|t|)") are lower than the predetermined significance level of 5%; thus, the data indicate that all three marketing instruments have a significant effect. For example, the value 0.013 in column "Pr(>|t|)" means that if we repeatedly draw samples from the entire population, the probability of not observing a relationship between the advertising budget and the quantity is 1.3%.

The column titled "Estimate" provides the values for the (rounded) regression coefficients; their standard errors are in the column "Std. Error." In contrast to other software programs, R does not automatically calculate the standardized beta values; therefore, separate calculations were necessary (values shown below the regression results in Fig. 2). In line with our expectations, person and the advertising budget have a positive impact, and price has a negative impact on quantity. Yet, we cannot use these regression coefficients to determine the strength of the effects. Instead, we must rely on the standardized beta values, which indicates that price has a slightly higher impact than the salesperson and that both marketing instruments show a considerably higher impact than advertising. Calculating the elasticities using Eq. 9,

we find an elasticity of the salespersons of 0.45, a price elasticity of -2.21 , and an advertising elasticity of 0.26. All elasticities show plausible signs and proportions (see also the meta-analyses conducted by Hanssens et al. 1990 and Albers et al.'s 2010 as well as Assmus et al. 1984; Bijmolt et al. 2005; Lodish et al. 1995; Sethuraman et al. 2011; Tellis 1988).

Assumptions

Thus far, we have focused on the method and the results of the linear regression. Next, we continue by examining whether the underlying statistical assumptions for a linear regression are fulfilled. These assumptions relate to the residuals; the relationship between dependent and independent variables, between the independent variable and the residuals, or between the independent variables; or the number of observations.

Assumptions for the residuals:

- Normal distribution of the residuals e_i
- Expected value of zero for the residuals $E(e_i) = 0$
- No correlation between residuals and independent variables, i.e., $\text{corr}(x_{i,k}, e_i) = 0$
- Constant variance of the residuals (homoscedasticity), i.e., $E(e_i^2) = \sigma^2$
- No correlation between the residuals (missing autocorrelation), i.e., $E(e_i \cdot e_i') = 0$

These assumptions essentially mean that the residuals resulting from the regression equation do not depend on the size of the observed variables (homoscedasticity), any of the independent variables, or the other residuals, particularly not on the residual of the previous period, as often occurs in the presence of time series (autocorrelation). In our example, the residual values should therefore be independent of the values of the three marketing instruments.

Endogeneity refers to the important assumption of the regression analysis that there is no relationship between one of the independent variables and the residuals. This independence should be present in our example because the company was naive in its marketing, which means that the company randomly chose the respective values for the marketing instruments. Therefore, no correlation should exist between the residuals, which reflect high or low level of marketing success, and the marketing instruments.

In reality, this assumption is often not fulfilled, because companies do not set their marketing activities randomly but systematically. These situations suffer from endogeneity. A typical example for such a situation occurs if the company does more marketing in districts where the company expects to be particularly successful. In our regression, “particularly successful” means that the quantity is higher than expected so that residuals are positive. Consequently, the independent variables correlate with the residuals.

Assumptions regarding the relationship between dependent and independent variables

- Consideration of all relevant independent variables
- Linearity of the regression equation (i.e., functional form)

To derive meaningful conclusions, we must ensure that the regression equation includes all relevant independent variables to avoid risking that the regression coefficients reflect the impact of the missing (also called omitted) variables. Furthermore, we assume in our linear regression analysis a linear relationship between the independent and dependent variables.

Assumptions regarding the relationship between the independent variables

- No multicollinearity between the independent variables

The regression analysis assumes that there is no linear relationship between the independent variables, that is, that there is a lack of multicollinearity. If multicollinearity occurs, because, for example, a high correlation exists between two independent variables, then a problem will occur: The effect of neither of the two variables on the dependent variable can be clearly derived.

Number of observations

- Sufficient number of observations

We can only estimate the regression equation in Eq. 1 if a sufficient number of observations are available. More precisely, the number of observations must be at least as large as the number of estimated coefficients (intercept b_0 plus the total number of regression coefficients b_k). Yet detecting a significant influence requires that the number of observations is much larger than the number of estimated coefficients. It is difficult to come up with a general statement about the ratio of the number of observations and the number of estimated coefficients, because this ratio always depends on the characteristics of variables in the data set. However, it is advisable that the number of observations is three times, better five times, as large as the number of estimated coefficients.

Procedure

Next, we describe the procedure for conducting a linear regression. We start by discussing the efficiency of the estimators. Afterwards, we discuss the problems that occur in case of multicollinearity, autocorrelation, heteroscedasticity, and outliers. In addition, we describe the transformation of variables often required to create a linear model.

Efficiency of Estimators

The estimated regression coefficient is efficient if it (1) is unbiased and (2) has the least variance of all unbiased coefficients (Gujarati 2003, p. 79). However, the least squares method only yields efficient estimates if the assumptions presented in section “[Assumptions](#)” hold. If the assumptions do not hold, then we must use a different estimation method or an alternative specification of the regression equation.

When examining the assumptions of the least squares method, we recommend investigating for the presence of multicollinearity, autocorrelation, and heteroscedasticity. If a sufficiently large number of observations are available, then the assumption of a normal distribution of the residuals is of minor importance, as it is usually fulfilled because of the central limit theorem (Stock and Watson 2015, p. 96). Koutsoyiannis (1977, p. 197) notes that this assumption is even fulfilled when samples have only 10–20 observations. In addition, the absence of a normal distribution signifies only that the F- and t-tests (discussed in section “[Significance Testing](#)”) are not meaningfully applicable. The estimated regression coefficients are still unbiased (Koutsoyiannis 1977, p. 197).

Test for Multicollinearity

Multicollinearity occurs when the independent variables are mutually linearly dependent. Multicollinearity usually leads to high standard errors of the estimated coefficients such that it is difficult to interpret them adequately. Inspecting the correlation matrix helps detect multicollinearity in the form of a linear dependency between two independent variables. High correlation values, often greater than -0.5 and $+0.5$, indicate multicollinearity, particularly if there are few observations. In this case, it is necessary to test the degree of the multicollinearity problem. We recommend running additional regressions that leave out some of the highly correlated variables. If the regression coefficients of the kept variables change substantially, multicollinearity represents a serious problem.

Running several linear regressions can help diagnose multicollinearity in the form of a linear dependence between more than two independent variables. In each regression, we use one of the original independent variables as a dependent variable and keep the others as independent variables. The difference between 1 and the R^2 for such a regression is called “tolerance,” and the inverse of this difference, which most statistical software programs report, is the “variance inflation factor” (VIF value). We can only assume a linear independence of the variables if the R^2 values of these regressions are low, which means that tolerance values are high, i.e., close to 1 and VIF values are low. Lower tolerance and higher VIF values, in contrast, indicate problems with multicollinearity. Although VIF values greater than 10 clearly indicate multicollinearity, even values of greater than 3, particularly with smaller data sets, point to problems with multicollinearity (see Hair et al. 2014, p. 200; for additional tests to uncover multicollinearity, see, e.g., Leeflang et al. 2000, pp. 348, 357).

Correlation Matrix			
	Person	Price	Advertising
Person	*****	0.143	-0.080
Price	0.597	*****	-0.158
Advertising	0.767	0.559	*****
upper diagonal part contains correlation coefficient estimates			
lower diagonal part contains corresponding p-values			

VIF-Values			
	Person	Price	Advertising
	1.024	1.044	1.029

Fig. 3 Testing for multicollinearity

In our example, the correlation matrix (see Fig. 3) indicates consistently low correlations between the independent variables, which shows there are no linear dependencies between two independent variables present. In addition, the VIF values close to 1 indicate that no linear dependencies between several independent variables exist.

With regard to the three marketing instruments, multicollinearity is thus not a problem for our data. Problems begin to emerge, however, if we also consider mailings (see Table 1), because the number of mailings is highly correlated (0.989) with the number of salespersons. Adding mailings to the linear regression analysis changes the results for the salespersons completely (Fig. 4); they now exert a negative influence on the quantity, while the number of mailings is highly positive. The high VIF values for the variables person and mailings also indicate the presence of multicollinearity.

An econometric solution for multicollinearity is difficult, although ridge regression (see Leeflang et al. 2000, p. 360) and partial least squares (Hair et al. 2017) might help. Increasing the number of observations can also help, in particular if the added observations exhibit a lower degree of linear dependence (here, salespersons and mailings). Alternatively, we could combine the linear dependent variables into a single variable (e.g., using factor analysis). We could also eliminate one or several of these variables from the linear regression analysis. For example, eliminating the salespersons from the regression decreases the parameter for mailings to 6.627 and its elasticity to 0.432.

Usually, however, it is difficult to find a truly satisfying solution for the multicollinearity problem because the number of observations is typically fixed and deleting variables is not helpful because the effects of these variables are often of primary interest. Therefore, in our numerical example, we can only assess the joint effect of both variables but not the specific effect of each variable, salespersons and mailings, on quantity. However, this information would certainly be of interest for the marketing manager in this case because this knowledge would enable her to optimize the marketing mix. For better determining the influence, the marketing manager should vary the number of salespersons and mailings so that a high correlation no longer occurs, ideally in the form of a field experiment.

```

Correlation Matrix
Person      Person  Price  Advertising  Mailings
Person      ***** 0.143 -0.080      0.989
Price       0.597 ***** -0.158      0.120
Advertising 0.767 0.559 ***** -0.115
Mailings    <0.001 0.659 0.672      *****

upper diagonal part contains correlation coefficient estimates
lower diagonal part contains corresponding p-values
-----

Model Summary

Call:
lm(formula = Quantity ~ Person + Price + Advertising + Mailings,
    data = regdata)

Residuals:
    Min       1Q   Median       3Q      Max
-5964.3 -3963.1   73.7   2805.2  5867.2

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 203009.252  23379.266   8.683  0.000 ***
Person      -1372.882   5815.151  -0.236  0.818
Price       -3711.737   435.531  -8.522  0.000 ***
Advertising    0.078     0.024   3.288  0.007 **
Mailings      7.949     5.654   1.406  0.187
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4372 on 11 degrees of freedom
Multiple R-squared:  0.9312,    Adjusted R-squared:  0.9062
F-statistic: 37.23 on 4 and 11 DF,  p-value: 2.473e-06
-----

VIF-Values
Person      Price      Advertising      Mailings
52.968      1.086      1.113      52.962
-----

Elasticities, at average values (not provided by R directly)
Person      Price      Advertising      Mailings
-0.091      -2.137      0.291      0.518

```

Fig. 4 Result of the regression analysis with multicollinearity problems

Test for Autocorrelation

Autocorrelation means that the residuals correlate with each other. Such correlation is most common in time series analyses, when the independent variables do not adequately cover the cyclical fluctuations in the time series. Autocorrelation usually leads to a situation where the predicted values are too high for some periods and too low for others. Thus, a series of negative residuals alternates with a series of positive residuals.

Autocorrelation causes an underestimation of the standard errors of the regression coefficients and, thus, an overestimation of the significance level of the t-test. (The standard error appears in the numerator of the t-test; see Eq. 7.) The estimated regression coefficients remain undistorted. However, they are no longer efficient because the standard error is not correctly detected (Wooldridge 2009, p. 408).

Most tests for autocorrelation examine the first-order autocorrelation, that is, the correlation between two consecutive residuals. In addition to graphical inspection of the residuals, the Durbin-Watson statistic (Wooldridge 2009, p. 415) is a common method to identify first-order autocorrelation:

$$dw = \frac{\sum_{i \in I'} (e_i - e_{i-1})^2}{\sum_{i \in I} e_i^2} \quad (10)$$

where

dw = value of the Durbin-Watson test and

I' = index set of observations without the first observation.

If the difference between the two successive residuals is very small (large), positive (negative) autocorrelation is present, and the numerator in Eq. 10 takes small (large) values. The Durbin-Watson value dw approaches 0 (4) in this situation. A value of 2 indicates that no first-order autocorrelation is present. For an accurate representation of the level of significance for the Durbin-Watson test, refer to Wooldridge (2009, p. 415) or Gujarati (2003, p. 970). Also note that tests for autocorrelation often make little sense for cross-sectional data unless the order of observations follows a particular trend.

In time series, the value of the dependent variable of the previous period often can serve as an independent variable, typically called the “lagged variable.” In this case, the Durbin-Watson statistic is not suitable for the detection of autocorrelation, and the Durbin h test should be used (see, e.g., Gujarati 2003, p. 503).

In our numerical example, considering autocorrelation is meaningless because we use cross-sectional data, in which temporal correlation cannot be present. At most, a spatial autocorrelation might be present (see Gujarati 2003, p. 442, for an explanation). Autocorrelation, however, can be a substantive problem because it is often an indication of missing (i.e., omitted) independent variables. Therefore, we illustrate the problem of autocorrelation with the numerical example in Table 2.

The columns with the x - and y -values describe the relationship between the x - and y -values in the following form:

$$y = 5 + 2 \times x. \quad (11)$$

The columns “True Error1” and “True Error2” provide the true residuals for two examples that illustrate first-order autocorrelation. These two residuals differ only by their signs. We compute the dependent variables y_1 and y_2 by adding the values of “True Error 1” and “True Error2” to the y -values, that is, $y_1 = y + \text{True Error1}$ and $y_2 = y + \text{True Error2}$.

Table 3 and Fig. 5 present the results of the linear regression analyses for the two numerical examples with the dependent variables y_1 and y_2 and the independent variable x . Note that the estimated residuals are systematically

Table 2 Numerical example to illustrate the problem of autocorrelation

Case	x	y	y1	y2	True Error1	True Error2
1	2	9	11	7	2	-2
2	3	11	15	7	4	-4
3	4	13	17	9	4	-4
4	5	15	17	13	2	-2
5	6	17	15	19	-2	2
6	7	19	15	23	-4	4
7	8	21	17	25	-4	4
8	9	23	21	25	-2	2

Table 3 Results of the linear regression analyses in the case of autocorrelation

	Regression 1 (with y1 as dep. var.)		Regression 2 (with y2 as dep. var.)	
	Intercept_1	Slope_1	Intercept_2	Slope_2
Value	11.286	0.857	-1.286	3.143
Standard error	1.882	0.316	1.882	0.316
Significance level t-test	0.00	0.03	0.52	0.00
R ²	0.55		0.94	
Significance level F-test	0.03		0.00	
Durbin-Watson value	1.27		1.27	

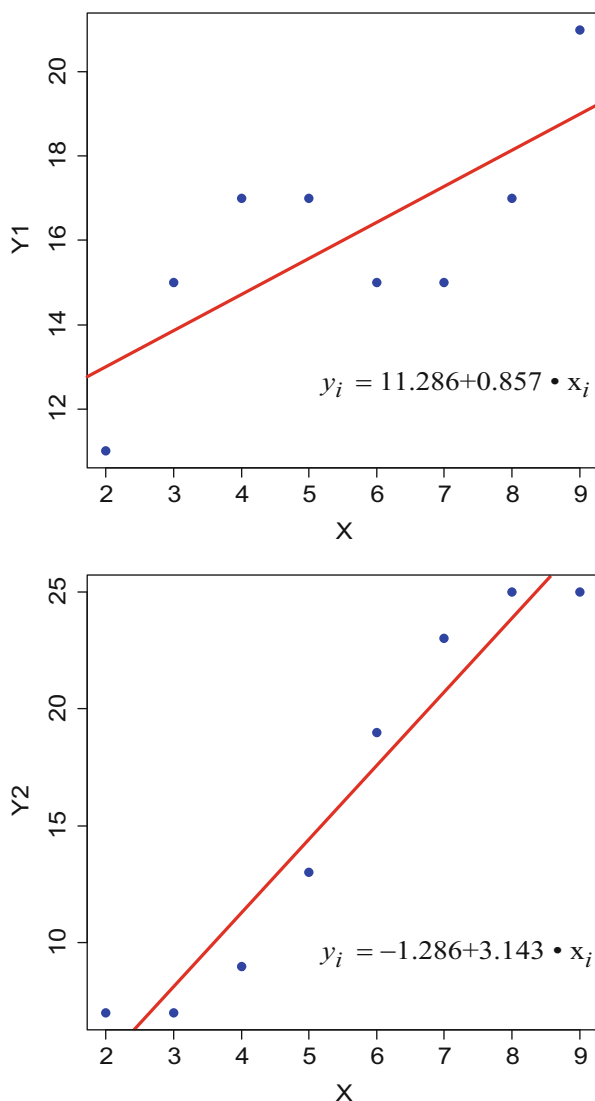
under- or overestimated in both examples and that the estimated regression equation does not reflect the actual functional relationship stated in Eq. 11. However, the estimated regression coefficients are still unbiased because the coefficients of Eq. 11 can be retained if we use a large number of random samples and run linear regression analyses on each of them. This result is also visible in Table 2 because the mean values of the constants and the regression coefficients of the two regression equations correspond to the coefficients in Eq. 11 ($\frac{11.29-1.29}{2} = 5$, respectively $\frac{0.86+3.14}{2} = 2$).

We can solve autocorrelation by detecting the factors responsible for the temporal fluctuations. Otherwise, we should consider econometric solutions such as the Cochrane-Orcutt procedure, the Hildreth-Lu procedure, or the Prais-Winsten estimate (Gujarati 2003, p. 482). These econometric methods, however, can only solve the problem if the regression equation is correctly specified.

Test for Heteroscedasticity

Heteroscedasticity means that the residuals do not have a constant variance. For example, the estimation of market shares for companies with high market shares might have a larger error than companies with small market shares (e.g., Gujarati 2003, p. 392). Heteroscedasticity leads to a situation in which the least squares method does not treat all observations equally and instead puts a greater emphasis on

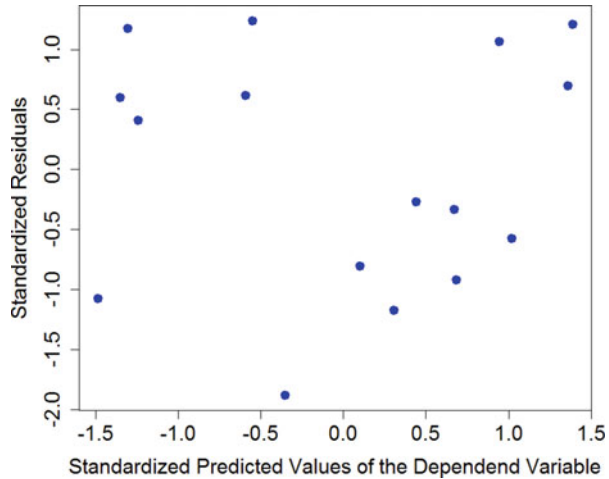
Fig. 5 Graphical representation of the regression equations in the case of the autocorrelation



the prediction of values with a high variance such that these observations receive a greater weight. The coefficients are still unbiased but are no longer efficient because they do not have the smallest estimation error (Stock and Watson 2015, p. 206).

We can detect heteroscedasticity by making a graphical comparison of the residuals with each of the independent variables or the dependent variable or by applying the Goldfeld-Quandt test, the Breusch-Pagan test, or the White test (Gujarati 2003, p. 400; Leeflang et al. 2000, p. 335). In our numerical example, the graphical comparison in Fig. 6 does not show a relationship between the standardized residuals and the standardized predicted values of the dependent variable, which indicates that heteroscedasticity is not present.

Fig. 6 Graphical inspection of heteroscedasticity



Unfortunately, heteroscedasticity is often not eliminated by collecting additional variables because the assumption of equal variances frequently does not make sense, as in our example of the estimation of market shares. It might help to transform variables in the regression equation (e.g., dividing the regression equation by the independent variable that causes the heteroscedasticity; Leeflang et al. 2000, p. 338). Alternatively, we can use a weighted linear regression analysis (“weighted least squares”) because it resolves the different weighting that results from heteroscedasticity (Gujarati 2003, pp. 415). When the data set consists of a large number of observations, calculating heteroscedastic robust standard error (Wooldridge 2009, p. 271) with procedures such as White robust standard errors (Gujarati 2003, pp. 439; White 1980) or the Newey-West estimator (Greene 2008, p. 643; Gujarati 2003, p. 439) also can be effective.

Identification of Outliers

The goal of linear regression analysis is usually that all observations have a comparable influence on the result; thus, it is optimal to avoid situations in which very few observations strongly influence the result. So-called outliers, whose values differ substantially from others in the data set, can cause such disproportional influence. Therefore, to examine for the presence of outliers, the easiest method is a visual inspection of the distribution of the observed values or the distribution of the residuals. In addition, a wide range of statistical methods are also effective. For example, the Mahalanobis distance is based on the standardized squared values of the independent variables, while Cook’s distance analyzes changes of the residuals that occur when the considered observation is removed from the regression equation. Chatterjee and Hadi (1986) provide a good overview of these and other statistical methods.

In our numerical example, the inspection of the residuals, the Mahalanobis distance, and the Cook’s distance do not show conspicuous values. However, district 17, which we have not yet used, has a much higher quantity than other districts. The

```

Model Summary

Call:
lm(formula = Quantity ~ Person + Price + Advertising, data =
regdata_wl7, x = TRUE, y = TRUE)

Residuals:
    Min       1Q   Median       3Q      Max
-14644  -5243  -1448   3443   29168

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 286804.698   51573.602    5.561  0.000 ***
Person       4294.625    1864.581    2.303   0.038 *
Price      -4710.844    1024.335   -4.599   0.000 ***
Advertising    0.009      0.054    0.159   0.876
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10840 on 13 degrees of freedom
Multiple R-squared:  0.6377,    Adjusted R-squared:  0.5541
F-statistic: 7.628 on 3 and 13 DF,  p-value: 0.003423

```

Fig. 7 Results of the linear regression analysis in the presence of an outlier

values of Cook's distance and Mahalanobis distance indicate that district 17 may represent an outlier. Including district 17 in our regression analysis causes a strong change in the coefficients and the significance levels compared with the previous results (Fig. 7). The corresponding elasticities for the number of salespersons, price, and advertising also change significantly (0.27, -2.64 , and 0.03 , respectively). In addition, the R^2 drops to 63.77%.

If the outliers are due to imputation errors, their values are easy to adjust. Otherwise, the solution is more difficult and depends largely on the question explored with the regression analysis. If the resulting recommendations should hold for all observations, eliminating outliers is not very satisfactory. Typically, however, recommendations are designed for the majority of the observations, so eliminating the outliers is appropriate.

In our example, we recommend eliminating the outlier (district 17) because the marketing manager is probably more interested in making recommendations that apply to the majority of the considered districts. However, at the same time, the marketing manager should carefully think about why district 17 is so different from the other districts. In any case, researchers should always carefully and precisely report which observations are considered outliers to prevent the appearance of data manipulation (Laurent 2013).

Transformation of Variables

When modeling quantity, it is frequently beneficial to estimate nonlinear relationships. The following equation is such a relationship that occurs in a multiplicative regression equation (i.e., a multiplicative sales response function):

$$Q = \alpha \cdot \text{Person}^\beta \cdot \text{Price}^\gamma \cdot \text{Advertising}^\delta \quad (12)$$

In contrast to the linear regression equation outlined in Eq. 1, the characteristics of this multiplicative regression equation are such that it captures interaction effects between the marketing instruments and varying marginal returns for each marketing instrument (see Gujarati 2003, p. 175 for a discussion of other functional forms).

It is still possible to use linear regression analysis to estimate such a nonlinear relationship, but to do so, transforming the variables so that a linear relationship between the independent variables and the dependent variable in the estimated regression equation is necessary. In the multiplicative regression equation, such a linear relation occurs if we take the logarithm of all variables:

$$\ln(Q) = \ln(\alpha) + \beta \cdot \ln(\text{Person}) + \gamma \cdot \ln(\text{Price}) + \delta \cdot \ln(\text{Advertising}). \quad (13)$$

The linear regression equation is then as follows:

$$\text{LN_}Q_i = \alpha' + \beta \cdot \text{LN_}Person_i + \gamma \cdot \text{LN_}P_i + \delta \cdot \text{LN_}AD_i + e_i \quad (i \in I), \quad (14)$$

where the variables LN_ Q_i , LN_ $Person_i$, LN_ P_i , and LN_ AD_i are defined as follows:

$$\begin{aligned} \text{LN_}Q_i &= \ln(Q_i), \\ \text{LN_}Person_i &= \ln(Person_i), \\ \text{LN_}P_i &= \ln(Price_i), \\ \text{LN_}AD_i &= \ln(Advertising_i), \text{ and} \\ \alpha' &= \ln(\alpha). \end{aligned}$$

We again ignore district 17. Figure 8 displays the results. Although the R^2 and the F-value are high, the R^2 only describes the goodness of fit of the logarithmic model shown in Eq. 14 and not that of the initial multiplicative model (Eq. 13). Therefore, calculating the R^2 of the initial model using the estimated values of the regression coefficients in Eq. 13 presents an alternative. The significance levels of all variables are comparable to those of the linear model. An important advantage of a multiplicative regression equation is that the regression coefficients represent the elasticities of the respective marketing instruments (Gujarati 2003, p. 176), which makes interpreting the results easier.

The results also show that all marketing instruments have the expected signs and plausible values. Inserting the estimated regression coefficients of Eq. 14 into Eq. 13 yields the following (rounded) results for the multiplicative regression equation:

$$\begin{aligned} Q &= \exp(16.98) \cdot \text{Person}^{0.40} \cdot \text{Price}^{-2.34} \cdot \text{Advertising}^{0.22} \\ &= 23,676,653 \cdot \text{Person}^{0.40} \cdot \text{Price}^{-2.34} \cdot \text{Advertising}^{0.22} \end{aligned} \quad (15)$$

Due to its more plausible properties, the multiplicative regression equation is more popular than the linear regression equation (see, e.g., Tellis 1988).

```

Call:
lm(formula = log(Quantity) ~ log(Person) + log(Price)
    + log(Advertising), data = regdata)

Residuals:
    Min       1Q   Median       3Q      Max
-0.10317 -0.02581  0.01209  0.03504  0.07761

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      16.981      1.493   11.372   0.000 ***
log(Person)         0.400      0.047    8.565   0.000 ***
log(Price)        -2.339      0.248   -9.440   0.000 ***
log(Advertising)   0.217      0.082    2.647   0.021 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.05198 on 12 degrees of freedom
Multiple R-squared:  0.9276,    Adjusted R-squared:  0.9095
F-statistic: 51.23 on 3 and 12 DF, p-value: 4.098e-07

```

Fig. 8 Results for the linear regression of the multiplicative regression equation

Implications of the Analysis

In this section, using our numerical example, we determine the optimal price and the amount of money that should be spend on advertising and salespersons. To do so, we rely on the estimated multiplicative regression equation (i.e., sales response function) shown in Eq. 16. We also ignore the marketing instrument mailings.

$$\text{Quantity} = 23,676,653 \cdot \text{Person}^{0.40} \cdot \text{Price}^{-2.34} \cdot \text{Advertising}^{0.22} \quad (16)$$

The profit function builds on this sales response function. Costs per unit are \$30 and the costs per salespersons are \$120,000 per year. Thus, profit is:

$$\text{Profit} = (\text{Price} - 30) \cdot 23,676,653 \cdot \text{Person}^{0.40} \cdot \text{Price}^{-2.34} \cdot \text{Advertising}^{0.22} - 120,000 \cdot \text{Person} - \text{Advertising} \quad (17)$$

Its derivation leads to:

$$\frac{\partial \text{Profit}}{\partial \text{Price}} = 23,676,653 \cdot \text{Person}^{0.40} \cdot (-1.34) \cdot \text{Price}^{-2.34} \cdot \text{Advertising}^{0.22} - 30 \cdot 23,676,653 \cdot \text{Person}^{0.40} \cdot (-2.34) \cdot \text{Price}^{-3.34} \cdot \text{Advertising}^{0.22} \quad (18)$$

$$\frac{\partial \text{Profit}}{\partial \text{Person}} = (\text{Price} - 30) \cdot 23,676,653 \cdot 0.40 \cdot \text{Person}^{-0.60} \cdot \text{Price}^{-2.34} \cdot \text{Advertising}^{0.22} - 120,000 \quad (19)$$

$$\frac{\partial \text{Profit}}{\partial \text{Advertising}} = (\text{Price} - 30) \cdot 23,676,653 \cdot \text{Person}^{0.40} \cdot \text{Price}^{-2.34} \cdot 0.22 \cdot \text{Advertising}^{-0.78} - 1 \quad (20)$$

By setting the derivatives to 0 and solving for the three marketing instruments, we can determine the optimal price and spending for salespersons as well as advertising budget. For our example, we obtain the following results (calculation with unrounded values):

$$\begin{aligned} \text{Price}^* &= 3652.40 \\ \text{Person}^* &= 36603,070.53 \\ \text{Advertising}^* &= 36327,402.06. \end{aligned}$$

The optimal value leads to (again calculating with unrounded values) a quantity of 67,398 units, revenue of \$3,531,459.94, profit contribution (i.e., profit before considering the budget for advertising and salespersons) of \$1,509,505.85, and profit of \$579,033.27.

Endogeneity

Endogeneity is currently one of the most important topics in linear regression analysis. Endogeneity means that one or more of the independent variables correlate with the residuals and therefore influences the relationship between the dependent variable and the independent variable. Its existence leads to a systematic distortion of the estimated regression coefficient. It can occur for several reasons, such as an omitted independent variable, simultaneity in the variables, measurement error in an independent variable, autocorrelation with delayed dependent variable, or self-selection.

We use our numerical example to discuss this fundamental problem. To do so, we first compute the residuals resulting from the linear regression analysis and the regression coefficients that are depicted in Fig. 2. We then add a column with these residuals to Table 1 and sort all districts according to the size of the residuals, which are defined as the difference between the observed and predicted (also called estimated) value of the dependent variable ($e_i = y_i - \hat{y}_i$). Positive residuals represent districts in which the company was more successful than an “average” district. Table 4 presents the results. For ease of exposition, we ignore the variable mailings and the outlier district 17 for the subsequent considerations.

A fundamental assumption of the linear regression analysis is that the correlation between each of the independent variables and the residuals is equal to 0 ($\text{corr}(x_{i,k}, u_i) = 0$). In line with our initial assumption that the company’s marketing is naive (i.e., random), this assumption appears realistic.

Now, more realistically, we assume that the company knows in which districts it is particularly successful and also selects its marketing instruments in such a way

Table 4 Districts with sales information sorted by the residuals

District	Quantity	Salespersons	Price	Advertising	Residuals
15	81,410	6	52	363,501	5,047.79
11	107,836	6	45	359,511	4,922.35
3	70,830	4	50	297,909	4,788.31
4	101,192	6	45	271,884	4,341.61
6	105,369	7	47	367,644	2,834.12
5	78,319	6	51	299,919	2,523.77
13	67,817	4	50	288,303	2,439.98
7	68,564	3	47	241,362	1,660.99
9	88,834	7	49	296,100	-1,085.46
2	91,735	5	46	370,062	-1,352.74
8	95,523	7	46	244,575	-2,328.75
1	81,996	7	49	228,753	-3,263.45
10	89,511	5	46	372,498	-3,745.29
14	59,207	6	54	289,470	-4,367.71
12	83,310	7	50	324,837	-4,765.38
16	71,431	3	46	361,974	-7,650.14

that it intensifies its marketing activities in the districts where it is already particularly successful. Thus, the company sets the highest of its 16 advertising budgets in the districts with the highest (positive) residuals, the second highest advertising budget in the district with the second highest residuals, and so on. We assume that the regression coefficients and residuals (see the results in Fig. 2) still apply and consider them the true parameters and the true residuals. Of course, given the newly allocated advertising budgets, the quantities in each district change. Yet, the total quantity of all 16 districts and the total advertising budget remain unchanged. However, the correlation between the true residuals (see Table 5) and the advertising budget is now 0.965.

Running a linear regression analysis with the data presented in Table 5 now leads to the results depicted in Table 6 (in the row “Endogeneity for advertising”). The coefficient for advertising is now more than twice as high (0.149 vs. 0.069; see Table 6). The reason for this increase in value is that the coefficient now reflects its own effect and the effect of the systematic allocation of the advertising budgets to districts with higher quantity. The estimate of the regression coefficient is therefore inconsistent and, thus, biased.

Endogeneity is difficult to detect because we do not observe the true residuals. Stated differently, the true relationship between advertising budget and the true residuals displayed in Table 5 with a correlation of 0.965 is unknown. We only observe the residuals that result from the linear regression analysis with the observations displayed in Table 5. The correlation of these residuals with the advertising budget is, by definition, 0.

The analogous procedure for the other two marketing instruments, price and salespersons, results in similar effects (see Table 6). Here, we assign the lowest

Table 5 Quantity of modified allocation of advertising budgets across districts

District	Quantity	Salespersons	Price	Advertising (endogenous)	True residuals
15	82,032.54	6	52	372,498	5,047.79
11	108,566.07	6	45	370,062	4,922.35
3	75,655.24	4	50	367,644	4,788.31
4	107,531.34	6	45	363,501	4,341.61
6	104,976.67	7	47	361,974	2,834.12
5	82,442.41	6	51	359,511	2,523.77
13	70,344.93	4	50	324,837	2,439.98
7	72,615.79	3	47	299,919	1,660.99
9	88,959.17	7	49	297,909	-1,085.46
2	86,617.28	5	46	296,100	-1,352.74
8	98,629.46	7	46	289,470	-2,328.75
1	86,116.50	7	49	288,303	-3,263.45
10	82,549.12	5	46	271,884	-3,745.29
14	56,100.54	6	54	244,575	-4,367.71
12	77,534.03	7	50	241,362	-4,765.38
16	62,212.91	3	46	228,753	-7,650.14

True regression equation:

Quantity = 210,159.444 + 6723.478 * Person - 3,832.503 * Price + 0.069 * Advertising + residual

Table 6 Regression results when endogeneity for the variables advertising, price, and salespersons is present

		Advertising	Price	Salespersons	Intercept
Without	Coefficient	0.069	-3,832.503	6,723.478	210,159.444
Endogeneity	Standard error	0.024	4,44.013	840.997	23,729.909
	p-value	0.013	0.000	0.000	0.000
	R ²	0.919			
Endogeneity	Coefficient	0.149	-3,733.320	6,381.400	182,421.558
For advertising	Standard error	0.005	101.972	195.945	5245.791
	p-value	0.000	0.000	0.000	0.000
	R ²	0.996			
Endogeneity	Coefficient	0.068	-5,318.478	6,579.367	283,245.887
For price	Standard error	0.004	73.352	140.682	3917.548
	p-value	0.000	0.000	0.000	0.000
	R ²	0.999			
Endogeneity	Coefficient	0.072	-3,679.197	9,460.590	186,682.532
For salespersons	Standard error	0.008	144.168	272.228	8040.702
	p-value	0.000	0.000	0.000	0.000
	R ²	0.995			

prices and highest number of salespersons to the district with the highest residuals, and so forth. The correlation between the true residuals and price is -0.984 , and between true residuals and the number of salespersons is 0.940 . We still assume that the other two independent variables have the values shown in Table 4.

Again, it is clear that the size and significance level of the regression coefficients increase substantially when the values of the respective variables are set systematically, that is, according to the quantities of the districts. Ebbes et al. (chapter ► [“Dealing with Endogeneity: A Nontechnical Guide for Marketing Researchers”](#)) provide a more detailed discussion of solutions in such situations.

Further Topics

As mentioned at the beginning of this chapter, linear regression analysis aims to relate a metric-scaled dependent variable with one or more metric-scaled independent variables. In practice, however, researchers often aim to include nonmetric information – for example, discrete variables such as gender, income groups, or shopping locations – or nonlinear (e.g., U-shaped) relationships.

Dummy variables are frequently used to include discrete information. For representing k characteristics, we require $k - 1$ dummy variables. The most common type of coding is indicator coding, in which each of the $k - 1$ dummy variables is represented by 1 (present) or 0 (not present). The k^{th} characteristic that is not represented by 1 is then the reference characteristic. The estimated $k - 1$ coefficients represent the difference of the respective characteristic with respect to the reference characteristic. We can thus interpret the coefficients as the difference in the intercepts of the characteristics. An alternative approach for the coding of dummy variables is so-called effect coding (for more details, refer to Hair et al. 2014, p. 173).

It is also possible to include nonlinear curve-shaped relationships in the regression by using appropriate transformation of the respective variables (Albers 2012). An example of such a relationship is a diminishing marginal effect of a marketing instrument such as advertising. In many situations, several types of data transformations are appropriate for linearizing a curvilinear relationship (Hair et al. 2014, pp. 174–175). Examples of direct approaches are arithmetic transformations such as taking the square root or the logarithm of a variable. Another method is the use of polynomials, most often, only the first (x_i) and second order (x_i^2) of the independent variables. Polynomials can help represent complex relationships, thereby making it possible to interpret and statistically test each coefficient individually. The independent variables would then describe the overall effect of the polynomial.

The discussion of nonlinear relationships thus far reflects the relationship of the dependent variable and an independent variable; however, situations in which a second independent variable influences the effect of an independent variable on the dependent variable, or a moderating or an interaction effect, are also common. To represent such an interaction effect, it is necessary to create a new variable by multiplying the

two independent variables. Thus, if two variables $x_{i,1}$ and $x_{i,2}$ interact – in other words, the variable $x_{i,2}$ moderates the influences of variable $x_{i,1}$ (or vice versa) – then we add the product of the two variables to the regression equation. The coefficient of these two variables ($x_{i,1} \cdot x_{i,2}$), also called the moderator, reflects the change in the effect of the variable $x_{i,1}$ if the variable $x_{i,2}$ changes. For a further discussion of the misspecification and interpretation of interaction effects, refer to Irwin and McClelland (2001).

For our numerical examples in this chapter, we use either cross-sectional (Table 1) or longitudinal (Table 2) data. However, many marketing studies use panel data (also called pooled data) (Wooldridge 2009, p. 10), which combines longitudinal and cross-sectional data. Their analysis requires the consideration of specific issues (e.g., the consideration of structural differences between the individual cross-sectional data) that are beyond the scope of this chapter. For further details, refer to Hsiao (2014).

Furthermore, researchers should consider whether the data have a hierarchical structure. In that case, hierarchical or multilevel regressions are recommended. Not accounting for a hierarchical data structure in turn leads to an error in the estimation of the coefficients. For a good introduction to the topic of multilevel modeling, see Snijders and Bosker (2012).

Software

Software for estimating a regression analysis is available in many forms. Most spreadsheet programs (e.g., Microsoft Excel) enable the calculation of some simple analyses, but these programs are limited with respect to graphical representation of the results as well as the number of statistical and econometric tests. In addition, they offer only a limited range of functions to transform data easily, examine different variants of the regression analysis, detect heteroscedasticity and autocorrelation, identify outliers, and apply nonlinear regression analysis.

More sophisticated programs such as SPSS (now owned by IBM) allow for performing these analyses easily. Although SPSS is easy to use, it does not include all methods for the detection and management of autocorrelation and heteroscedasticity (e.g., the Durbin's h statistic). Other statistical programs such as SAS, STATA, EViews, and LIMDEP provide a higher functionality, although they are slightly less user-friendly. Furthermore, none of these programs are available free of charge. An open source program, PSPP, is intended as an alternative to IBM SPSS. In principle, matrix-oriented programs such as R, MATLAB, or Gauss also allow for applying all econometric methods because the user works directly with matrices. However, these programs are less easy to use and have a considerably greater learning curve. R's advantage is that it is open source and available free of charge. In summary, most spreadsheet programs are sufficient for the occasional computation of regression analysis, but for more detailed analyses, researchers should use one of the aforementioned statistical or matrix-oriented programs.

Summary

We use a numerical example to describe in detail one of the most important statistical methods, linear regression analysis. It examines the linear relationship between a dependent variable and one or more independent variables but can be easily used to also estimate nonlinear relationships if they can be linearized, as is the case for a multiplicative sales response functions (and many other functions). We use an extensive numerical example to illustrate all aspects that we cover in this section (autocorrelation, multicollinearity, heteroscedasticity, outlier detection, endogeneity, optimization of marketing mix), and we provide the code (for R, SPSS, and STATA) that we used to perform all calculations on our website (www.skiera.de). In addition, we also provide an Excel spreadsheet that contains all calculations, which should help in even better understanding of all calculations. Yet, we would like to highlight that Excel is not a good environment to conduct these calculations.

References

- Albers, S. (2012). Optimizable and implementable aggregate response modeling for marketing decision support. *International Journal of Research in Marketing*, 29(2), 111–122.
- Albers, S., Mantrala, M. K., & Sridhar, S. (2010). Personal selling elasticities: A meta-analysis. *Journal of Marketing Research*, 47(5), 840–853.
- Assmus, G., Farley, J. W., & Lehmann, D. R. (1984). How advertising affects sales: A meta-analysis of econometric results. *Journal of Marketing Research*, 21(1), 65–74.
- Bijmolt, T. H. A., van Heerde, H., & Pieters, R. G. M. (2005). New empirical generalizations on the determinants of price elasticity. *Journal of Marketing Research*, 42(2), 141–156.
- Chatterjee, S., & Hadi, A. S. (1986). Influential observations, high leverage points, and outliers in linear regressions. *Statistical Science*, 1(3), 379–416.
- Greene, W. H. (2008). *Econometric analysis* (6th ed.). Upper Saddle River: Pearson.
- Gujarati, D. N. (2003). *Basic econometrics* (4th ed.). New York: McGraw Hill.
- Hair, J. F., Black, W. C., Babin, J. B., & Anderson, R. E. (2014). *Multivariate data analysis* (7th ed.). Upper Saddle River: Pearson.
- Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2017). *A primer on partial least squares structural equation modeling (PLS-SEM)* (2nd ed.). Thousand Oaks: Sage.
- Hanssens, D. M., Parsons, L. J., & Schultz, R. L. (1990). *Market response models: Econometric and time series analysis*. Boston: Springer.
- Hsiao, C. (2014). *Analysis of panel data* (3rd ed.). Cambridge: Cambridge University Press.
- Irwin, J. R., & McClelland, G. H. (2001). Misleading heuristics and moderated multiple regression models. *Journal of Marketing Research*, 38(1), 100–109.
- Koutsoyiannis, A. (1977). *Theory of econometrics* (2nd ed.). Houndmills: MacMillan.
- Laurent, G. (2013). EMAC distinguished marketing scholar 2012: Respect the data! *International Journal of Research in Marketing*, 30(4), 323–334.
- Leeflang, P. S. H., Wittink, D. R., Wedel, M., & Neart, P. A. (2000). *Building models for marketing decisions*. Berlin: Kluwer.
- Lodish, L. L., Abraham, M. M., Kalmenson, S., Livelsberger, J., Lubetkin, B., Richardson, B., & Stevens, M. E. (1995). How TV advertising works: A meta-analysis of 389 real world split cable T. V. advertising experiments. *Journal of Marketing Research*, 32(2), 125–139.
- Pindyck, R. S., & Rubinfeld, D. (1998). *Econometric models and econometric forecasts* (4th ed.). New York: McGraw-Hill.

- Sethuraman, R., Tellis, G. J., & Briesch, R. A. (2011). How well does advertising work? Generalizations from meta-analysis of brand advertising elasticities. *Journal of Marketing Research*, 48(3), 457–471.
- Snijders, T. A. B., & Bosker, R. J. (2012). *Multilevel analysis: An introduction to basic and advanced multilevel modeling* (2nd ed.). London: Sage.
- Stock, J., & Watson, M. (2015). *Introduction to econometrics* (3rd ed.). Upper Saddle River: Pearson.
- Tellis, G. J. (1988). The price sensitivity of selective demand: A meta-analysis of econometric models of sales. *Journal of Marketing Research*, 25(4), 391–404.
- White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4), 817–838.
- Wooldridge, J. M. (2009). *Introductory econometrics: A modern approach* (4th ed.). Mason: South-Western Cengage.